

Carburettor Icing – Multiple Choice Questions

Section 1 – General Question

1.

What is carburettor icing?

- A. Fuel contamination in the tanks
- B. Ice forming inside the carburettor restricting airflow ☒
- C. Ice on the propeller
- D. Ice on the wings

Answer: B

2.

Carburettor icing is most common in:

- A. Jet aircraft
- B. Turbine engines
- C. Light piston-engine aircraft ☒
- D. Gliders only

Answer: C

3.

Carb ice can result in:

- A. Increased engine power
- B. Reduced power or engine failure ☒
- C. Faster climb performance
- D. Better fuel economy

Answer: B

4.

Carb ice forms because of:

- A. Exhaust heat
- B. Rapid cooling inside the carburettor ☒
- C. Electrical failure
- D. Magnetic interference

Answer: B

5.

The Venturi effect causes:

- A. Increased pressure
- B. Increased temperature
- C. Reduced pressure and temperature ☒
- D. Fuel contamination

Answer: C

6.

Fuel evaporation inside the carburettor:

- A. Heats the air
- B. Cools the air further ☒
- C. Has no effect
- D. Increases pressure

Answer: B

7.

A temperature drop inside the carburettor may be:

- A. 1–5°C
- B. 5–10°C
- C. 15–30°C ☒
- D. 40–50°C

Answer: C

8.
Moisture freezes primarily on the:
A. Spark plugs
B. Exhaust manifold
C. Venturi and throttle valve ☒
D. Magnetos
Answer: C

9.
Carb ice most commonly forms around:
A. Fuel tank cap
B. Throttle butterfly valve ☒
C. Propeller hub
D. Oil filter
Answer: B

10.
Carburettor icing can occur:
A. Only below 0°C
B. Only in snow
C. Even on warm, humid days ☒
D. Only in winter
Answer: C

11.
Fuel-injected engines are generally not susceptible to:
A. Engine overheating
B. Carburettor icing ☒
C. Oil leaks
D. Magneto failure
Answer: B

12.
Fuel-injected engines can still experience:
A. Induction icing ☒
B. Carburettor icing
C. Wing icing only
D. Fuel vaporisation
Answer: A

13.
The highest-risk temperature range for carb icing is approximately:
A. -20°C to -10°C
B. +5°C to +20°C ☒
C. +25°C to +35°C
D. +40°C to +50°C
Answer: B

14.
High relative humidity:
A. Reduces carb icing risk
B. Has no effect
C. Increases carb icing risk ☒
D. Prevents icing
Answer: C

15.
Visible moisture includes:
A. Fog ☒
B. Sunlight
C. Turbulence
D. Wind shear
Answer: A

16.

A small temperature-dew point spread indicates:

- A. Dry air
- B. Saturated air with high humidity ✓
- C. Strong winds
- D. Low pressure

Answer: B

17.

Carb ice is most likely during:

- A. Full-power climb
- B. Engine start
- C. Low-power operation ✓
- D. Cruise at high power

Answer: C

18.

A prolonged descent can encourage:

- A. Carb icing ✓
- B. Detonation
- C. Compression loss
- D. Static discharge

Answer: A

19.

One of the first signs of carb ice in a fixed-pitch prop aircraft is:

- A. Oil pressure increase
- B. Gradual RPM drop ✓
- C. Fuel exhaustion
- D. CHT increase

Answer: B

20.

In a constant-speed propeller aircraft, carb ice is often indicated by:

- A. Oil temperature rise
- B. RPM rise
- C. Drop in manifold pressure ✓
- D. Radio interference

Answer: C

21.

Engine rough running may indicate:

- A. Carb ice ✓
- B. Good engine performance
- C. Fuel richness correction
- D. Normal operation

Answer: A

22.

An unresponsive throttle may indicate:

- A. Excessive fuel pressure
- B. Ice restricting airflow ✓
- C. Good power output
- D. Magneto failure

Answer: B

23.

A physically stiff throttle might occur because:

- A. The cable stretched
- B. Ice formed around it ✓
- C. Fuel froze
- D. Battery is low

Answer: B

24.

An unexplained loss of engine power should make a pilot suspect:

- A. Carb icing ✓
- B. Better efficiency
- C. Improved mixture
- D. Strong tailwind

Answer: A

25.

The best strategy for carb ice is:

- A. Ignore it
- B. Prevent it ✓
- C. Wait for it to melt
- D. Shutdown immediately

Answer: B

26.

Before flight, pilots should review:

- A. Humidity and temperature ✓
- B. Passenger weight only
- C. Airport coffee prices
- D. Radio frequencies only

Answer: A

27.

During run-up, carb heat should cause:

- A. Slight RPM drop ✓
- B. Large RPM increase
- C. No change
- D. Engine shutdown

Answer: A

28.

Typical RPM drop during carb heat check:

- A. 10 RPM
- B. 50–150 RPM ✓
- C. 500 RPM
- D. 1000 RPM

Answer: B

29.

A non-functioning carb heat system means:

- A. Greater icing risk ✓
- B. Better performance
- C. More range
- D. Lower fuel burn

Answer: A

30.

Pilots should limit:

- A. High power operations
- B. Long periods of idle power in humid conditions ✓
- C. Cruise flight
- D. Straight-and-level flight

Answer: B

31.

Many training aircraft require carb heat during:

- A. Takeoff roll
- B. Landing approach ✓
- C. Initial climb
- D. Parking

Answer: B

32.

Applying carb heat periodically in humid cruise conditions can:

- A. Detect and remove developing ice ✓
- B. Increase speed
- C. Improve radio quality
- D. Prevent oil leaks

Answer: A

33.

When using carb heat:

- A. Full ON or OFF ✓
- B. Half ON only
- C. Quarter ON
- D. Any position

Answer: A

34.

Partial carb heat can:

- A. Improve efficiency
- B. Worsen icing ✓
- C. Melt all ice instantly
- D. Increase RPM

Answer: B

35.

The first action when carb ice is suspected is:

- A. Switch tanks
- B. Apply full carb heat ✓
- C. Turn off mixture
- D. Reduce power

Answer: B

36.

Apply carb heat:

- A. Gradually only
- B. Fully and immediately ✓
- C. After landing
- D. Only above 5000 ft

Answer: B

37.

Carb heat works by directing:

- A. Cold outside air
- B. Warm air into the carburettor ✓
- C. Oil vapour
- D. Hydraulic pressure

Answer: B

38.

After applying carb heat, pilots should:

- A. Stop flying the aircraft
- B. Maintain aircraft control ✓
- C. Turn off radios
- D. Shut down engine

Answer: B

39.

The engine may initially run:

- A. Smoother immediately
- B. Rougher while ice melts ✓
- C. Faster instantly
- D. Hotter only

Answer: B

40.

Initial roughness after carb heat often means:

- A. Ice is melting ☒
- B. Engine failure is certain
- C. Carb heat is broken
- D. Oil pressure is low

Answer: A

41.

If roughness increases after carb heat:

- A. Turn heat off immediately
- B. Keep heat ON ☒
- C. Shut engine down
- D. Lean aggressively

Answer: B

42.

Power should:

- A. Gradually recover ☒
- B. Disappear permanently
- C. Increase beyond normal
- D. Stay unchanged

Answer: A

43.

Power recovery may take:

- A. A few seconds to several minutes ☒
- B. 30 minutes
- C. 1 hour
- D. A full day

Answer: A

44.

After recovery, carb heat should be:

- A. Left on forever
- B. Gradually returned to OFF ☒
- C. Removed from aircraft
- D. Ignored

Answer: B

45.

If power does not recover:

- A. Ignore the problem
- B. Prepare for an emergency landing ☒
- C. Continue normally
- D. Increase speed

Answer: B

46.

No improvement after several minutes may indicate:

- A. Another problem exists ☒
- B. Better fuel economy
- C. Improved engine health
- D. Normal operation

Answer: A

47.

A common misconception is:

- A. Carb ice occurs in warm weather
- B. Carb ice only occurs in cold weather ☒
- C. Carb heat melts ice
- D. Humidity affects icing

Answer: B

48.

Most carb ice incidents occur:

- A. Well above freezing ☒
- B. Below -20°C only
- C. In Arctic regions only
- D. Only in snowfall

Answer: A

49.

Carb ice can form:

- A. In clear air ☒
- B. Only in rain
- C. Only in clouds
- D. Only in hail

Answer: A

50.

High humidity alone can:

- A. Promote carb ice ☒
- B. Prevent carb ice
- C. Stop fuel evaporation
- D. Raise RPM

Answer: A

51.

Using intermittent carb heat improperly can:

- A. Improve climb performance
- B. Cause refreezing ☒
- C. Eliminate all ice
- D. Increase power

Answer: B

52.

One cause of accidents is:

- A. Leaving carb heat on long enough
- B. Turning carb heat off too soon ☒
- C. Checking carb heat on run-up
- D. Monitoring RPM

Answer: B

53.

During a go-around, carb heat is usually:

- A. Left on indefinitely
- B. Turned off for full power ☒
- C. Removed
- D. Disconnected

Answer: B

54.

Heated air is:

- A. More dense
- B. Less dense ☒
- C. Equal density
- D. Frozen

Answer: B

55.

Leaving carb heat on reduces:

- A. Drag
- B. Available power ☒
- C. Fuel flow
- D. Altitude

Answer: B

56.

Failure to check carb heat before takeoff can:

- A. Increase risk ✓
- B. Improve safety
- C. Prevent icing
- D. Increase lift

Answer: A

57.

A good preflight habit is to check:

- A. Carb heat operation ✓
- B. Paint condition only
- C. Passenger bags
- D. Cabin lights only

Answer: A

58.

The best defense against carb ice is:

- A. Awareness ✓
- B. Luck
- C. Speed only
- D. Altitude only

Answer: A

59.

Carb heat is usually operated using:

- A. Full ON or OFF positions ✓
- B. Tiny adjustments only
- C. Automatic settings
- D. Fuel selector

Answer: A

60.

Carburettor icing can affect:

- A. Fuel-air mixture ✓
- B. Landing gear only
- C. Wings only
- D. Rudder only

Answer: A

61.

Ice restricts:

- A. Airflow ✓
- B. Cabin ventilation only
- C. Electrical current
- D. Hydraulic fluid

Answer: A

62.

Reduced airflow causes:

- A. Power loss ✓
- B. Better climb rate
- C. Longer engine life
- D. Increased RPM

Answer: A

63.

Carb ice can occur during taxi because:

- A. Low power settings encourage it ✓
- B. The propeller stops
- C. Oil freezes
- D. Mixture is always lean

Answer: A

64.

The throttle plate is also called the:

- A. Venturi
- B. Butterfly valve ☒
- C. Mixture needle
- D. Injector

Answer: B

65.

The carburettor mixes:

- A. Oil and fuel
- B. Air and fuel ☒
- C. Water and fuel
- D. Fuel and electricity

Answer: B

66.

Moist air and cooling are the key ingredients for:

- A. Carb icing ☒
- B. Detonation
- C. Vapor lock
- D. Overspeed

Answer: A

67.

A carburetted engine commonly found in trainers may be:

- A. Lycoming O-320 ☒
- B. Turbofan engine
- C. Turbojet engine
- D. Rotor engine only

Answer: A

68.

The carburettor throat is also called the:

- A. Exhaust stack
- B. Venturi ☒
- C. Spinner
- D. Intake scoop

Answer: B

69.

A pilot should suspect carb ice if RPM drops:

- A. Without changing throttle ☒
- B. During shutdown only
- C. After refueling
- D. During maintenance

Answer: A

70.

Engine vibration with power loss may indicate:

- A. Carb icing ☒
- B. Better engine tuning
- C. Correct mixture
- D. Low humidity

Answer: A

71.

Carb icing is considered:

- A. A common engine icing hazard ☒
- B. Impossible
- C. Very rare
- D. Unimportant

Answer: A

72.

The main recovery tool for carb icing is:

- A. Carb heat ☒
- B. Brakes
- C. Flaps
- D. Trim

Answer: A

73.

Aviation priority during recovery is:

- A. Aviate first ☒
- B. Communicate first
- C. Navigate first
- D. Troubleshoot first

Answer: A

74.

Warm humid air can be:

- A. Ideal for carb icing ☒
- B. Completely safe from icing
- C. Too hot for icing
- D. Irrelevant

Answer: A

75.

High humidity above approximately 50%:

- A. Increases risk ☒
- B. Eliminates risk
- C. Freezes fuel
- D. Reduces cooling

Answer: A

76.

The worst carb icing conditions often involve:

- A. High humidity and moderate temperature ☒
- B. Dry desert air
- C. Strong winds only
- D. Night flying only

Answer: A

77.

Carb icing may contribute to:

- A. Engine failure ☒
- B. Increased thrust
- C. Better fuel flow
- D. Higher RPM

Answer: A

78.

Pilots should apply carb heat before:

- A. Extended low-power descents ☒
- B. Engine shutdown only
- C. Refueling
- D. Taxi completion

Answer: A

79.

If carb heat is inoperative:

- A. Icing cannot be cleared effectively ☒
- B. Ice cannot form
- C. Performance improves
- D. Run-up is unnecessary

Answer: A

80.

The safest approach to carb icing is:

- A. Wait until the engine stops
- B. Apply carb heat only after landing
- C. Recognize early and act immediately ✓
- D. Ignore small RPM changes

Answer: C

81.

Carburettor ice forms most readily when:

- A. Air contains moisture ✓
- B. Air is completely dry
- C. The engine is off
- D. Fuel tanks are empty

Answer: A

82.

The Venturi effect causes air to:

- A. Slow down and warm up
- B. Speed up and cool down ✓
- C. Stop moving
- D. Increase pressure

Answer: B

83.

When carb heat is selected ON, the engine initially loses some power because:

- A. Warm air is less dense ✓
- B. Fuel flow stops
- C. Oil pressure drops
- D. Magnetos switch off

Answer: A

84.

Carburettor icing is classified as:

- A. Structural icing
- B. Engine induction icing ✓
- C. Instrument icing
- D. Airframe icing

Answer: B

85.

Which phase of flight is especially vulnerable to carb icing?

- A. Extended descent ✓
- B. Full-power climb
- C. Takeoff roll only
- D. Engine shutdown

Answer: A

86.

The carb heat source is usually warmed by:

- A. Battery power
- B. Exhaust system heat ✓
- C. Hydraulic pressure
- D. Wing heaters

Answer: B

87.

If carb ice is suspected, carb heat should remain on until:

- A. The engine stops
- B. Ice has melted and power is restored ✓
- C. ATC instructs otherwise
- D. Landing is completed

Answer: B

88.

A pilot notices a gradual RPM decrease and rough running. The most likely cause is:

- A. Carb icing ☒
- B. Tailwind increase
- C. Better fuel economy
- D. Static electricity

Answer: A

89.

Applying carb heat introduces:

- A. Hot, unfiltered air in many aircraft ☒
- B. Cold outside air
- C. Extra fuel
- D. Oxygen only

Answer: A

90.

Carburettor icing can occur at temperatures as high as:

- A. 30°C or higher in humid conditions ☒
- B. 0°C only
- C. -15°C only
- D. -40°C only

Answer: A

91.

The purpose of a carb heat check during run-up is to:

- A. Verify the system operates correctly ☒
- B. Increase RPM
- C. Check magnetos
- D. Lean the mixture

Answer: A

92.

Carb ice is dangerous because it can:

- A. Restrict airflow to the engine ☒
- B. Increase lift
- C. Reduce drag
- D. Improve fuel flow

Answer: A

93.

A severe carb icing condition may eventually cause:

- A. Engine failure ☒
- B. Higher manifold pressure
- C. Extra thrust
- D. Better performance

Answer: A

94.

When carb heat melts ice, the engine may ingest:

- A. Water from melted ice ☒
- B. Hydraulic fluid
- C. Engine oil
- D. Brake fluid

Answer: A

95.

Which condition produces the highest carb icing risk?

- A. Warm humid day ☒
- B. Cold dry day
- C. Hot desert afternoon
- D. Very low humidity

Answer: A

96.

A pilot should monitor which indication in a fixed-pitch propeller aircraft?

- A. RPM ☒
- B. Cabin temperature only
- C. Brake pressure only
- D. Voltage only

Answer: A

97.

In a constant-speed propeller aircraft, carb icing is usually first noticed through:

- A. Reduced manifold pressure ☒
- B. Higher RPM
- C. Increased oil pressure
- D. Lower fuel quantity

Answer: A

98.

Carb heat should normally be selected:

- A. Fully ON or fully OFF ☒
- B. One-quarter ON
- C. Half ON permanently
- D. Randomly adjusted

Answer: A

99.

The key to avoiding carb icing incidents is:

- A. Early recognition and prompt action ☒
- B. Flying faster
- C. Reducing fuel load
- D. Climbing higher

Answer: A

100.

The most effective management of carburettor icing is:

- A. Prevention, detection, and correct carb heat use ☒
- B. Ignoring minor symptoms
- C. Shutting down the engine immediately
- D. Relying on luck

Answer: A

101.

Carburettor icing is most likely when air is:

- A. Moist and cooling rapidly ☒
- B. Completely dry
- C. Very hot and dry
- D. Oxygen-free

Answer: A

102.

The carburettor's Venturi causes air pressure to:

- A. Increase
- B. Decrease ☒
- C. Stay the same
- D. Become unstable only at high altitude

Answer: B

103.

A drop in pressure inside the carburettor causes air temperature to:

- A. Rise
- B. Fall ☒
- C. Stay constant
- D. Become hotter than exhaust gas

Answer: B

104.

Fuel evaporation contributes to carb icing because it:

- A. Adds heat
- B. Removes heat from the air ☒
- C. Increases oil temperature
- D. Prevents moisture from freezing

Answer: B

105.

Carb ice can begin forming when the outside air temperature is:

- A. Only below freezing
- B. Above freezing if humidity is high ☒
- C. Only below -20°C
- D. Only during snow

Answer: B

106.

A major danger of carb ice is that it may develop:

- A. Suddenly with no possible symptoms
- B. Gradually and unnoticed ☒
- C. Only after engine shutdown
- D. Only during refuelling

Answer: B

107.

In a fixed-pitch propeller aircraft, carb icing usually causes:

- A. RPM decrease ☒
- B. RPM increase
- C. Oil pressure increase
- D. Airspeed increase

Answer: A

108.

In an aircraft with a constant-speed propeller, carb icing often causes:

- A. Manifold pressure decrease ☒
- B. Manifold pressure increase
- C. Propeller overspeed
- D. Fuel quantity increase

Answer: A

109.

Carb heat should be applied when:

- A. Carb ice is suspected ☒
- B. The engine is producing perfect power
- C. The aircraft is parked
- D. The master switch is off

Answer: A

110.

The correct carb heat action for suspected icing is:

- A. Full carb heat ON ☒
- B. Half carb heat ON
- C. Carb heat OFF
- D. Mixture idle cut-off

Answer: A

111.

After applying carb heat, engine roughness may increase because:

- A. Ice is melting and passing through the engine ☒
- B. The engine is improving immediately
- C. The propeller has stopped
- D. The fuel tanks are empty

Answer: A

112.

If rough running occurs after carb heat is selected, the pilot should usually:

- A. Leave carb heat ON until ice clears ✓
- B. Turn carb heat OFF immediately
- C. Shut the engine down
- D. Close the throttle fully

Answer: A

113.

Carb heat uses air that is:

- A. Heated before entering the carburettor ✓
- B. Cooled before entering the carburettor
- C. Mixed with hydraulic fluid
- D. Taken from the fuel tank

Answer: A

114.

Using carb heat usually causes a slight:

- A. RPM drop ✓
- B. RPM rise
- C. Fuel quantity increase
- D. Oil level increase

Answer: A

115.

The slight RPM drop during a carb heat check indicates:

- A. Warm air is entering the carburettor ✓
- B. The magnetos have failed
- C. The brakes are working
- D. The fuel selector is closed

Answer: A

116.

No RPM drop during a carb heat check may indicate:

- A. Carb heat may not be functioning properly ✓
- B. The aircraft is overloaded
- C. The tyres are underinflated
- D. The altimeter is incorrect

Answer: A

117.

Carb icing is usually associated with:

- A. Moisture in the air ✓
- B. Completely dry air only
- C. Empty fuel tanks
- D. Low oil quantity only

Answer: A

118.

The throttle butterfly valve is a common place for:

- A. Ice accumulation ✓
- B. Tyre wear
- C. Compass error
- D. Radio interference

Answer: A

119.

Carb ice restricts airflow, causing the mixture to become:

- A. Richer ✓
- B. Leaner only
- C. Unchanged
- D. Pure oxygen

Answer: A

120.

A richer mixture caused by carb icing may lead to:

- A. Rough engine running ✓
- B. Higher lift
- C. Reduced drag
- D. Stronger brakes

Answer: A

121.

Which is a common symptom of carb icing?

- A. Gradual power loss ✓
- B. Increased indicated airspeed
- C. Improved climb rate
- D. Reduced aircraft weight

Answer: A

122.

Carb icing is especially hazardous because it can lead to:

- A. Complete engine failure ✓
- B. Increased fuel capacity
- C. Better cruise performance
- D. More electrical power

Answer: A

123.

A pilot should be especially alert for carb ice during:

- A. Low-power operation ✓
- B. Full-power takeoff only
- C. Engine shutdown only
- D. Parking brake checks

Answer: A

124.

Long descents at reduced power increase carb icing risk because:

- A. The throttle is partly closed ✓
- B. The tyres cool down
- C. The battery charges faster
- D. The altimeter stops working

Answer: A

125.

Before a prolonged descent, pilots often apply:

- A. Carb heat ✓
- B. Parking brake
- C. Pitot cover
- D. Master switch OFF

Answer: A

126.

During approach in many carburetted aircraft, carb heat is used because:

- A. Power is reduced and icing risk increases ✓
- B. The aircraft is accelerating rapidly
- C. The engine is shut down
- D. The fuel tanks are empty

Answer: A

127.

Carb heat should normally be removed before full power is required because:

- A. Heated air reduces engine power ✓
- B. It increases lift too much
- C. It blocks the brakes
- D. It turns off the magnetos

Answer: A

128.

During a go-around, carb heat is usually:

- A. Selected OFF as power is applied ✓
- B. Left ON forever
- C. Pulled halfway out
- D. Used instead of throttle

Answer: A

129.

Warm air entering the carburettor is less dense, so engine power:

- A. Decreases slightly ✓
- B. Increases greatly
- C. Becomes unlimited
- D. Is unaffected in all aircraft

Answer: A

130.

A correct response to suspected carb ice is:

- A. Aviate, apply carb heat, monitor engine ✓
- B. Panic and shut down immediately
- C. Ignore it until landing
- D. Reduce airspeed below stall

Answer: A

131.

Carb icing may occur:

- A. In clear air ✓
- B. Only inside clouds
- C. Only in thunderstorms
- D. Only on icy runways

Answer: A

132.

Visible moisture is:

- A. Fog, cloud, rain, or mist ✓
- B. Dry wind only
- C. Clear desert air
- D. Cabin heat

Answer: A

133.

A small temperature-dew point spread usually means:

- A. High humidity ✓
- B. Very dry air
- C. No carb icing possible
- D. Strong tailwind

Answer: A

134.

The closer the temperature is to the dew point, the:

- A. Higher the moisture content is likely to be ✓
- B. Lower the humidity is likely to be
- C. Less likely cloud is
- D. More fuel is available

Answer: A

135.

Carb icing can happen during taxi because:

- A. Low power settings may encourage icing ✓
- B. The aircraft is flying too fast
- C. The engine is stopped
- D. The wings are producing maximum lift

Answer: A

136.

If carb ice forms on the throttle plate, it can:

- A. Restrict throttle movement ☒
- B. Increase fuel quantity
- C. Inflate tyres
- D. Improve radio range

Answer: A

137.

A stiff or unresponsive throttle may indicate:

- A. Ice near the throttle valve ☒
- B. Too much runway length
- C. Correct engine operation
- D. Excessive cabin ventilation

Answer: A

138.

If carb icing is suspected and power continues to reduce, the pilot should:

- A. Prepare for possible forced landing ☒
- B. Ignore the symptoms
- C. Turn off carb heat immediately
- D. Close the fuel selector

Answer: A

139.

Carb heat is part of the aircraft's:

- A. Induction system protection ☒
- B. Landing gear system
- C. Navigation system
- D. Brake system only

Answer: A

140.

Carburettor icing mainly affects the:

- A. Engine air intake and fuel-air mixture ☒
- B. Elevator trim tab
- C. Main landing gear
- D. Flap hinges

Answer: A

141.

The carburettor's job is to mix:

- A. Fuel and air ☒
- B. Oil and water
- C. Battery acid and fuel
- D. Hydraulic fluid and oxygen

Answer: A

142.

When ice forms inside the carburettor, the engine receives:

- A. Less air ☒
- B. More clean air
- C. More electrical current
- D. More oil pressure

Answer: A

143.

Less air entering the engine usually causes:

- A. Reduced power ☒
- B. Increased lift
- C. Better braking
- D. Higher fuel capacity

Answer: A

144.

Carb icing risk is increased by:

- A. Humidity ✓
- B. Dry air
- C. Empty seats
- D. Strong sunlight only

Answer: A

145.

Carb icing is least likely in:

- A. Very dry air ✓
- B. Moist air
- C. Fog
- D. Humid conditions

Answer: A

146.

The carburettor temperature can be much lower than:

- A. Outside air temperature ✓
- B. Compass heading
- C. Aircraft registration
- D. Runway elevation only

Answer: A

147.

Carb ice can form even when outside air temperature is above 0°C because:

- A. Cooling inside the carburettor can reduce temperature below freezing ✓
- B. The wings are cold
- C. The tyres are cold
- D. The radio is switched on

Answer: A

148.

The most important pilot action when engine performance decreases unexpectedly is to:

- A. Consider carb ice as a possible cause ✓
- B. Ignore it
- C. Increase cabin heat
- D. Switch off avionics

Answer: A

149.

A gradual RPM drop in cruise without throttle movement may suggest:

- A. Carb icing ✓
- B. Stronger engine performance
- C. Increased fuel capacity
- D. Better mixture control

Answer: A

150.

If carb heat clears ice successfully, the pilot should expect:

- A. Power recovery ✓
- B. Immediate engine shutdown
- C. Permanent RPM loss
- D. Total electrical failure

Answer: A

151.

If carb icing has been severe, power recovery may:

- A. Take some time ✓
- B. Always happen instantly
- C. Never occur
- D. Require tyre pressure changes

Answer: A

152.

Carb heat should not be turned off too early because:

- A. Ice may remain or reform ☒
- B. The aircraft will stop flying
- C. The brakes will fail
- D. The compass will freeze

Answer: A

153.

The correct use of carb heat requires:

- A. Patience while ice melts ☒
- B. Instant removal after roughness starts
- C. Avoiding it during all descents
- D. Using only partial heat

Answer: A

154.

Partial carb heat is generally avoided because:

- A. It may not fully clear ice and can worsen conditions ☒
- B. It always increases power
- C. It stops fuel flow
- D. It disables the propeller

Answer: A

155.

The safest carb heat setting when treating ice is:

- A. Full ON ☒
- B. One-quarter ON
- C. Half ON
- D. Random position

Answer: A

156.

The safest carb heat setting when not needed is usually:

- A. Full OFF ☒
- B. Half ON
- C. Slightly ON
- D. Fully open throttle only

Answer: A

157.

A carb heat control is commonly located:

- A. In the cockpit ☒
- B. On the wingtip only
- C. Inside the fuel tank
- D. On the tailplane

Answer: A

158.

Carb icing is more likely at reduced power because:

- A. The throttle plate is more closed ☒
- B. The propeller stops
- C. The aircraft is lighter
- D. Fuel disappears

Answer: A

159.

A closed or partly closed throttle plate increases ice risk because:

- A. It creates a restriction where ice can collect ☒
- B. It warms the air
- C. It removes moisture
- D. It increases fuel tank pressure

Answer: A

160.

When carb ice blocks airflow, the engine may:

- A. Run rough ☒
- B. Become quieter with more power
- C. Produce more thrust
- D. Use no fuel

Answer: A

161.

Which aircraft system is directly involved in carb icing?

- A. Induction system ☒
- B. Landing light system
- C. Seatbelt system
- D. Static wick system

Answer: A

162.

Which condition should make a pilot more alert for carb ice?

- A. Humid air and reduced power ☒
- B. Dry air and full power
- C. Empty baggage area
- D. Smooth runway

Answer: A

163.

A pilot should check carb heat during:

- A. Engine run-up ☒
- B. Passenger briefing only
- C. Wheel chock removal only
- D. After shutdown only

Answer: A

164.

During run-up, applying carb heat checks that:

- A. Heated air can enter the carburettor ☒
- B. The flaps are symmetrical
- C. The tyres are inflated
- D. The radio works

Answer: A

165.

If carb heat is applied and RPM rises after an initial drop, this may indicate:

- A. Ice was present and is clearing ☒
- B. The brakes are working
- C. The altimeter is accurate
- D. The aircraft is overweight

Answer: A

166.

A rise in RPM after carb heat has been on for a short time can mean:

- A. Ice has melted and airflow improved ☒
- B. Fuel tanks are overfilled
- C. Magnetos are off
- D. Oil pressure is zero

Answer: A

167.

If carb heat is ineffective and the engine continues losing power, the pilot should:

- A. Follow engine failure procedures ☒
- B. Continue as normal
- C. Turn off the fuel
- D. Close all vents only

Answer: A

168.

In carb icing conditions, situational awareness includes monitoring:

- A. RPM or manifold pressure ✓
- B. Paint colour
- C. Seat position only
- D. Passenger snacks

Answer: A

169.

A pilot should be cautious of carb icing during:

- A. Training circuits with repeated power reductions ✓
- B. Only high-speed cruise
- C. Only refuelling
- D. Only parking

Answer: A

170.

Repeated approaches and descents can increase exposure to:

- A. Carb icing conditions ✓
- B. Better fuel economy only
- C. Radio failure only
- D. Compass repair

Answer: A

171.

Which is the best description of carb icing?

- A. Ice forming inside the carburettor due to cooling and moisture ✓
- B. Ice forming only on the windshield
- C. Fuel freezing inside the tank
- D. Oil freezing in the sump only

Answer: A

172.

The main cause of the cooling that leads to carb ice is:

- A. Pressure drop and fuel evaporation ✓
- B. Cabin ventilation
- C. Brake friction
- D. Electrical resistance

Answer: A

173.

Carb ice may form even when there is no airframe icing because:

- A. Carburettor temperature can be lower than outside air ✓
- B. Airframe icing always comes first
- C. Wings are warmer than the engine
- D. The pitot tube controls carb temperature

Answer: A

174.

A pilot should not assume carb icing is impossible just because:

- A. The outside temperature is above freezing ✓
- B. The aircraft is moving
- C. The runway is paved
- D. The radios work

Answer: A

175.

The presence of mist or haze may suggest:

- A. Moist air and increased carb icing risk ✓
- B. Dry air and no risk
- C. Empty fuel tanks
- D. Increased tyre pressure

Answer: A

176.

The best way to confirm carb heat works before takeoff is:

- A. Apply it during run-up and observe expected RPM drop ✓
- B. Ignore it unless ice forms
- C. Open the door
- D. Check the landing light

Answer: A

177.

If carb heat is selected ON during normal cruise, the pilot should expect:

- A. A small power decrease ✓
- B. A large automatic climb
- C. No engine change ever
- D. Flaps to extend

Answer: A

178.

Carb heat air may be unfiltered, so pilots usually avoid using it:

- A. On dusty ground unless necessary ✓
- B. In the air at all times
- C. During an emergency
- D. During suspected icing

Answer: A

179.

Carb ice prevention includes:

- A. Using carb heat as recommended in the aircraft checklist ✓
- B. Never checking carb heat
- C. Flying only at night
- D. Keeping the mixture rich at all times

Answer: A

180.

The aircraft flight manual or POH should be used for:

- A. Specific carb heat procedures ✓
- B. Guessing runway length
- C. Ignoring engine symptoms
- D. Replacing training

Answer: A

181.

If carb icing occurs during descent, the pilot should:

- A. Apply carb heat and monitor engine response ✓
- B. Close the throttle completely and wait
- C. Turn off the magnetos
- D. Switch off the fuel selector

Answer: A

182.

Carb icing is dangerous on final approach because:

- A. Power may be needed quickly ✓
- B. The aircraft cannot turn
- C. The brakes stop working in the air
- D. The radio becomes louder

Answer: A

183.

If carb ice is present on final approach, applying full power may:

- A. Not produce expected power ✓
- B. Always produce excess power
- C. Stop all airflow
- D. Extend the flaps automatically

Answer: A

184.

Carb heat should be considered before reducing power because:

- A. Ice may form during low-power operation ✓
- B. The tyres will cool
- C. The runway will disappear
- D. The fuel gauges stop working

Answer: A

185.

Which condition is most associated with carb icing?

- A. High humidity ✓
- B. Very dry desert air
- C. Empty cockpit
- D. High tyre pressure

Answer: A

186.

Carb icing risk can exist during:

- A. Spring, summer, autumn, or winter ✓
- B. Winter only
- C. Snow only
- D. Night only

Answer: A

187.

A common mistake is to believe carb icing happens only:

- A. In freezing outside air temperatures ✓
- B. In humid air
- C. In descent
- D. At reduced power

Answer: A

188.

Carb heat selected ON normally bypasses:

- A. Normal cold air intake route ✓
- B. The fuel tanks
- C. The magnetos
- D. The oil pump

Answer: A

189.

The warm air used for carb heat commonly comes from around the:

- A. Exhaust ✓
- B. Fuel tank cap
- C. Altimeter
- D. Wingtip

Answer: A

190.

When carb heat is ON, mixture may become:

- A. Richer because warm air is less dense ✓
- B. Leaner because fuel stops
- C. Pure oxygen
- D. Unchanged in all cases

Answer: A

191.

A richer mixture from carb heat can cause:

- A. Slight roughness or RPM drop ✓
- B. Immediate tyre failure
- C. Better radio reception
- D. Increased wing lift

Answer: A

192.

Carb icing symptoms should be treated:

- A. Promptly ☒
- B. Only after landing
- C. Only during maintenance
- D. After the engine stops

Answer: A

193.

Engine roughness plus power loss in humid conditions should make a pilot suspect:

- A. Carburettor icing ☒
- B. Improved engine efficiency
- C. Lower aircraft weight
- D. Better propeller performance

Answer: A

194.

The main pilot control used to remove carb ice is:

- A. Carb heat ☒
- B. Rudder trim
- C. Landing light
- D. Transponder

Answer: A

195.

If carb icing is suspected, the pilot should avoid:

- A. Delaying carb heat application ☒
- B. Maintaining aircraft control
- C. Monitoring instruments
- D. Preparing for emergency actions if needed

Answer: A

196.

Carb ice can reduce engine power by:

- A. Blocking airflow through the carburettor ☒
- B. Increasing runway length
- C. Lowering aircraft weight
- D. Improving combustion

Answer: A

197.

A useful habit in carb-ice-prone conditions is to:

- A. Regularly monitor engine indications ☒
- B. Ignore small power changes
- C. Avoid checklists
- D. Use partial carb heat always

Answer: A

198.

Carburettor icing is best managed through:

- A. Awareness, prevention, recognition, and correct action ☒
- B. Waiting for it to disappear
- C. Flying without checking weather
- D. Avoiding engine instruments

Answer: A

199.

If engine power is restored after carb heat, the pilot should:

- A. Continue to monitor for re-icing ☒
- B. Ignore the engine from then on
- C. Shut down immediately
- D. Turn off all instruments

Answer: A

200.

The safest pilot attitude toward carb icing is:

- A. Expect it when conditions are suitable and act early 
- B. Assume it cannot happen in warm weather
- C. Use carb heat only after engine failure
- D. Ignore gradual RPM loss

Answer: A

Section 2 - Chart Questions

Multiple-choice questions based on the attached Carburettor Icing Chart-Master, which shows a carburettor icing-probability chart using temperature, dew point, dew point depression, relative humidity, and icing-risk zones.

1. What is the main purpose of the carburettor icing-probability chart?

- A. To calculate aircraft fuel consumption
- B. To determine the probability of carburettor icing
- C. To measure engine oil temperature
- D. To calculate aircraft weight and balance

Correct answer: B. To determine the probability of carburettor icing

Detailed answer:

The chart is specifically labelled as a carburettor icing-probability chart. It is used to estimate the likelihood and severity of carburettor icing by comparing air temperature with dew point depression and interpreting the resulting position against the shaded icing-risk areas.

2. Which two weather values must first be obtained before using the chart?

- A. Wind speed and wind direction
- B. Temperature and pressure altitude
- C. Temperature and dew point
- D. Cloud base and visibility

Correct answer: C. Temperature and dew point

Detailed answer:

The chart instructions state that to use the chart, the pilot should first obtain the temperature and dew point. These two values are needed to calculate the dew point depression, which is then used on the chart.

3. What is meant by “dew point depression” on this chart?

- A. The difference between cloud base and freezing level
- B. The difference between temperature and dew point
- C. The difference between sea-level pressure and station pressure
- D. The difference between cruise power and descent power

Correct answer: B. The difference between temperature and dew point

Detailed answer:

The chart explains that dew point depression is calculated by finding the difference between the air temperature and the dew point. For example, if the temperature is 12°C and the dew point is 2°C, the dew point depression is 10°C.

4. If the temperature is 12°C and the dew point is 2°C, what is the dew point depression?

- A. 2°C
- B. 10°C
- C. 12°C
- D. 14°C

Correct answer: B. 10°C

Detailed answer:

Dew point depression is calculated by subtracting the dew point from the air temperature. In this case, 12°C – 2°C = 10°C. This exact example is provided in the chart instructions.

5. On the chart, what does the horizontal axis represent?

- A. Relative humidity
- B. Dew point depression
- C. Temperature in degrees Celsius
- D. Engine power setting

Correct answer: C. Temperature in degrees Celsius

Detailed answer:

The bottom axis of the chart is labelled Temperature °C. This axis is used to locate the outside air temperature before comparing it with the dew point depression.

6. What does the vertical axis on the left side of the chart represent?
- A. Pressure altitude
 - B. Dew point depression
 - C. Carburettor temperature
 - D. Relative humidity

Correct answer: B. Dew point depression

Detailed answer:

The left-hand vertical scale shows values for dew point depression. The chart uses this value together with the temperature to locate the appropriate point within an icing-risk zone.

7. What does the right-hand scale of the chart refer to?
- A. Fuel pressure
 - B. Relative humidity
 - C. Engine RPM
 - D. Maximum altitude

Correct answer: B. Relative humidity

Detailed answer:

The chart instructions state that for relative humidity, the user should refer to the right-hand scale. This means the chart also provides a humidity reference alongside the icing-probability zones.

8. How is icing probability determined on the chart?
- A. By checking aircraft speed against altitude
 - B. By finding where temperature and dew point depression intersect
 - C. By measuring fuel flow and air pressure
 - D. By comparing wind speed with runway direction

Correct answer: B. By finding where temperature and dew point depression intersect

Detailed answer:

The chart instructions explain that for icing probability, the user should refer to the shading legend appropriate to the intersection of the lines. This means the point where the temperature value and dew point depression value meet determines the likely icing risk.

9. Which of the following is one of the icing-risk categories shown on the chart?
- A. Severe turbulence
 - B. Moderate icing at cruise power
 - C. Hydraulic icing
 - D. Wingtip frost only

Correct answer: B. Moderate icing at cruise power

Detailed answer:

One of the chart's labelled risk zones is moderate icing at cruise power. The chart also includes other zones such as serious icing at descent power and light icing at cruise or descent power.

10. Which power setting is specifically associated with serious icing in one of the chart zones?
- A. Take-off power
 - B. Climb power only
 - C. Descent power
 - D. Idle cut-off

Correct answer: C. Descent power

Detailed answer:

The chart includes a labelled zone for serious icing at descent power. This indicates that carburettor icing risk can be especially significant during lower-power operations such as descent.

11. Which statement best describes how to begin using the chart?

- A. First calculate aircraft centre of gravity
- B. First obtain temperature and dew point
- C. First select engine RPM
- D. First determine magnetic heading

Correct answer: B. First obtain temperature and dew point

Detailed answer:

The chart's "To use this chart" instructions begin by telling the user to obtain the temperature and dew point. These are the starting values needed before calculating dew point depression.

12. If the air temperature is 20°C and the dew point is 15°C, what is the dew point depression?

- A. 5°C
- B. 15°C
- C. 20°C
- D. 35°C

Correct answer: A. 5°C

Detailed answer:

Dew point depression is calculated as temperature minus dew point. Therefore, $20^{\circ}\text{C} - 15^{\circ}\text{C} = 5^{\circ}\text{C}$. A small dew point depression generally means the air temperature is close to the dew point, which corresponds to higher humidity conditions on this type of chart.

13. If the temperature is 25°C and the dew point is 10°C, what is the dew point depression?

- A. 10°C
- B. 15°C
- C. 25°C
- D. 35°C

Correct answer: B. 15°C

Detailed answer:

Subtract the dew point from the temperature: $25^{\circ}\text{C} - 10^{\circ}\text{C} = 15^{\circ}\text{C}$. This value would then be used on the chart's dew point depression scale to locate the icing-probability zone.

14. What does a smaller dew point depression indicate?

- A. The air is drier
- B. The temperature is much higher than the dew point
- C. The air temperature is closer to the dew point
- D. The aircraft is flying faster

Correct answer: C. The air temperature is closer to the dew point

Detailed answer:

Dew point depression is the difference between temperature and dew point. A smaller difference means the two values are closer together. On the chart, this is relevant because dew point depression is one of the key inputs used to assess icing probability.

15. Which of the following combinations would produce the greatest dew point depression?

- A. Temperature 10°C, dew point 9°C
- B. Temperature 15°C, dew point 10°C
- C. Temperature 25°C, dew point 5°C
- D. Temperature 12°C, dew point 2°C

Correct answer: C. Temperature 25°C, dew point 5°C

Detailed answer:

Calculate each difference:

- $10^{\circ}\text{C} - 9^{\circ}\text{C} = 1^{\circ}\text{C}$
- $15^{\circ}\text{C} - 10^{\circ}\text{C} = 5^{\circ}\text{C}$
- $25^{\circ}\text{C} - 5^{\circ}\text{C} = 20^{\circ}\text{C}$
- $12^{\circ}\text{C} - 2^{\circ}\text{C} = 10^{\circ}\text{C}$

The largest dew point depression is therefore 20°C, from the 25°C and 5°C combination.

16. Which chart feature tells the user the severity of carburettor icing risk?

- A. The aircraft registration
- B. The shaded legend or shaded zones
- C. The runway length scale
- D. The compass rose

Correct answer: B. The shaded legend or shaded zones

Detailed answer:

The chart instructions say that icing probability is determined by referring to the shading legend at the intersection of the relevant temperature and dew point depression lines. The shaded areas represent different icing-risk categories.

17. Which of the following is NOT shown as a main input for using the chart?

- A. Temperature
- B. Dew point
- C. Dew point depression
- D. Aircraft registration

Correct answer: D. Aircraft registration

Detailed answer:

The chart uses temperature, dew point, and dew point depression to assess carburettor icing probability. Aircraft registration is not part of the chart's calculation or interpretation process.

18. What should a pilot do after calculating dew point depression?

- A. Ignore the temperature scale
- B. Locate the intersection of temperature and dew point depression on the chart
- C. Convert the aircraft weight into pounds
- D. Calculate runway slope

Correct answer: B. Locate the intersection of temperature and dew point depression on the chart

Detailed answer:

After calculating dew point depression, the next step is to use the chart by matching the temperature with the dew point depression. The intersection of these values falls within a shaded zone that indicates the icing probability.

19. Which risk description appears on the chart together with cruise or descent power?

- A. Light icing
- B. Structural overload
- C. Fuel starvation
- D. Magneto failure

Correct answer: A. Light icing

Detailed answer:

The chart includes a labelled area for light icing at cruise or descent power. This shows that different levels of icing risk may apply depending on atmospheric conditions and engine power setting.

20. Why is this chart useful for pilots operating carburettor-equipped aircraft?

- A. It helps estimate icing risk before or during flight
- B. It gives the aircraft's exact fuel endurance
- C. It replaces the need for pre-flight weather information
- D. It calculates take-off distance automatically

Correct answer: A. It helps estimate icing risk before or during flight

Detailed answer:

The chart helps pilots assess carburettor icing probability using temperature, dew point, dew point depression, relative humidity reference, and shaded icing-risk areas. This makes it useful for anticipating conditions where carburettor icing may occur.

Section 3

Part A — Coastal Regions: Scenario Questions

1. Cape Town coastal morning

A student pilot at Cape Town notes:

- Temperature: 16°C
- Dew point: 14°C

What is the dew point depression, and what does it suggest when using the chart?

- A. 30°C — very dry air
- B. 14°C — low humidity
- C. 2°C — moist air and increased icing concern
- D. 16°C — no need to use the chart

Correct answer: C. 2°C — moist air and increased icing concern

Detailed answer:

Dew point depression = temperature – dew point.

So, 16°C – 14°C = 2°C. A small dew point depression means the air temperature is close to the dew point. On the carburettor icing chart, this value is plotted against temperature to identify the relevant icing-risk zone. The chart specifically instructs the user to calculate dew point depression and then use the intersection of the temperature and dew point depression lines.

2. Durban humid coastal afternoon

Near Durban, the conditions are:

- Temperature: 26°C
- Dew point: 22°C

What should the pilot calculate first?

- A. Pressure altitude
- B. Dew point depression
- C. Wind correction angle
- D. Density altitude only

Correct answer: B. Dew point depression

Detailed answer:

The first calculation required by the chart is dew point depression:

26°C – 22°C = 4°C.

The chart is used by obtaining temperature and dew point, calculating the difference, and then referring to the shaded legend at the intersection point.

3. Gqeberha coastal training flight

The reported weather is:

- Temperature: 20°C
- Dew point: 15°C

What is the dew point depression?

- A. 35°C
- B. 20°C
- C. 15°C
- D. 5°C

Correct answer: D. 5°C

Detailed answer:

Dew point depression = 20°C – 15°C = 5°C.

This is the value to use on the chart's left-hand vertical scale, while 20°C is found on the bottom temperature scale. The intersection tells the user which shaded icing-probability area applies.

4. George/Garden Route low cloud scenario

A pilot preparing for a coastal flight near George has:

- Temperature: 18°C
- Dew point: 18°C

What is the dew point depression?

- A. 0°C
- B. 18°C
- C. 36°C
- D. 9°C

Correct answer: A. 0°C

Detailed answer:

$$18^{\circ}\text{C} - 18^{\circ}\text{C} = 0^{\circ}\text{C}.$$

A dew point depression of 0°C means the air is saturated or very close to saturation. On the chart, the pilot should plot 18°C on the temperature axis and 0°C on the dew point depression axis, then check the shaded icing-risk region.

5. East London coastal departure

East London conditions:

- Temperature: 23°C
- Dew point: 19°C

Which calculation is correct?

- A. $23 + 19 = 42^{\circ}\text{C}$ dew point depression
- B. $23 - 19 = 4^{\circ}\text{C}$ dew point depression
- C. $19 - 23 = -4^{\circ}\text{C}$ dew point depression
- D. $23 \div 19 = 1.2^{\circ}\text{C}$ dew point depression

Correct answer: B. $23 - 19 = 4^{\circ}\text{C}$ dew point depression

Detailed answer:

The chart defines dew point depression as the difference between temperature and dew point.

Therefore:

$$23^{\circ}\text{C} - 19^{\circ}\text{C} = 4^{\circ}\text{C}.$$

The pilot would then use temperature 23°C and dew point depression 4°C to find the intersection on the chart.

6. Richards Bay humid coast

A pilot operating from Richards Bay observes:

- Temperature: 28°C
- Dew point: 24°C

Which statement is most correct?

- A. Dew point depression is 4°C, so the chart must still be checked
- B. Dew point depression is 52°C, so no icing is possible
- C. Dew point depression is 24°C, so only altitude matters
- D. Dew point depression is 28°C, so the aircraft type is irrelevant

Correct answer: A. Dew point depression is 4°C, so the chart must still be checked

Detailed answer:

$$\text{Dew point depression} = 28^{\circ}\text{C} - 24^{\circ}\text{C} = 4^{\circ}\text{C}.$$

Although the temperature is warm, the attached chart is designed to assess icing probability by combining temperature and dew point depression, not temperature alone.

7. Mossel Bay cool coastal morning

Conditions are:

- Temperature: 12°C
- Dew point: 9°C

What is the dew point depression?

- A. 3°C
- B. 9°C
- C. 12°C
- D. 21°C

Correct answer: A. 3°C

Detailed answer:

$$12^{\circ}\text{C} - 9^{\circ}\text{C} = 3^{\circ}\text{C}.$$

This small depression indicates moist air. On the chart, the pilot should locate 12°C along the temperature axis and 3°C on the dew point depression axis, then read the shaded icing-risk zone.

8. Cape south coast descent

During descent along the south coast, the reported values are:

- Temperature: 14°C
- Dew point: 7°C

What is the dew point depression?

- A. 21°C
- B. 7°C
- C. 14°C
- D. 0°C

Correct answer: B. 7°C

Detailed answer:

$$\text{Dew point depression} = 14^{\circ}\text{C} - 7^{\circ}\text{C} = 7^{\circ}\text{C}.$$

Because one of the chart's labelled risk categories refers to serious icing at descent power, pilots should be especially careful interpreting the chart during descent or low-power operation.

9. KwaZulu-Natal coastal cruise

A coastal cruise near KwaZulu-Natal has:

- Temperature: 30°C
- Dew point: 20°C

What is the correct dew point depression?

- A. 10°C
- B. 20°C
- C. 30°C
- D. 50°C

Correct answer: A. 10°C

Detailed answer:

$$30^{\circ}\text{C} - 20^{\circ}\text{C} = 10^{\circ}\text{C}.$$

The temperature value goes on the chart's bottom scale, while the dew point depression goes on the left-hand scale. The intersection determines the icing-probability zone.

10. Coastal comparison question

Which coastal scenario has the smallest dew point depression and therefore the moistest air?

- A. Cape Town: 16°C / 14°C
- B. Durban: 26°C / 22°C
- C. Gqeberha: 20°C / 15°C
- D. KwaZulu-Natal coast: 30°C / 20°C

Correct answer: A. Cape Town: 16°C / 14°C

Detailed answer:

Calculate each:

- Cape Town: $16 - 14 = 2^{\circ}\text{C}$
- Durban: $26 - 22 = 4^{\circ}\text{C}$
- Gqeberha: $20 - 15 = 5^{\circ}\text{C}$
- KwaZulu-Natal coast: $30 - 20 = 10^{\circ}\text{C}$

The smallest dew point depression is 2°C, meaning the temperature is closest to the dew point. The chart uses this dew point depression value as one of its two main plotting inputs.

Part B — Inland Regions: Scenario Questions

11. Johannesburg Highveld summer afternoon

Johannesburg has:

- Temperature: 27°C
- Dew point: 12°C

What is the dew point depression?

- A. 12°C
- B. 15°C
- C. 27°C
- D. 39°C

Correct answer: B. 15°C

Detailed answer:

$$27^{\circ}\text{C} - 12^{\circ}\text{C} = 15^{\circ}\text{C}.$$

The inland plateau/Highveld commonly experiences different conditions from the coast because altitude strongly affects temperature patterns in South Africa.

For the chart, the pilot uses 27°C and 15°C dew point depression to locate the intersection.

12. Bloemfontein winter morning

Conditions:

- Temperature: 8°C
- Dew point: 2°C

What is the correct dew point depression?

- A. 6°C
- B. 8°C
- C. 10°C
- D. 2°C

Correct answer: A. 6°C

Detailed answer:

$$8^{\circ}\text{C} - 2^{\circ}\text{C} = 6^{\circ}\text{C}.$$

The value 6°C is plotted on the left-hand dew point depression scale, while 8°C is plotted on the bottom temperature scale. The chart's shaded zone at their intersection provides the icing-probability interpretation.

13. Kimberley dry inland heat

Kimberley reports:

- Temperature: 34°C
- Dew point: 8°C

What is the dew point depression?

- A. 8°C
- B. 26°C
- C. 34°C
- D. 42°C

Correct answer: B. 26°C

Detailed answer:

$$34^{\circ}\text{C} - 8^{\circ}\text{C} = 26^{\circ}\text{C}.$$

This is a large dew point depression, meaning the air is much drier than in many coastal examples. South Africa's inland and north-western regions can be much drier than coastal regions.

The pilot should still use the chart method rather than judging by temperature alone.

14. Upington hot inland day

Upington values:

- Temperature: 36°C
- Dew point: 10°C

What is the dew point depression?

- A. 10°C
- B. 26°C
- C. 36°C
- D. 46°C

Correct answer: B. 26°C

Detailed answer:

$$36^{\circ}\text{C} - 10^{\circ}\text{C} = 26^{\circ}\text{C}.$$

This indicates relatively dry air. However, the correct use of the chart is still to plot the temperature and dew point depression, then read the shading legend at the intersection.

15. Nelspruit/Mbombela humid inland-lowveld scenario

Conditions near Mbombela:

- Temperature: 25°C
- Dew point: 20°C

What is the dew point depression?

- A. 5°C
- B. 20°C
- C. 25°C
- D. 45°C

Correct answer: A. 5°C

Detailed answer:

$$25^{\circ}\text{C} - 20^{\circ}\text{C} = 5^{\circ}\text{C}.$$

Although this is an inland scenario, it has a small dew point depression, so it should not be dismissed as “dry inland air.” The chart requires the actual temperature and dew point depression values to be plotted.

16. Polokwane dry inland morning

Polokwane conditions:

- Temperature: 22°C
- Dew point: 7°C

What is the correct calculation?

- A. $22 - 7 = 15^{\circ}\text{C}$
- B. $22 + 7 = 29^{\circ}\text{C}$
- C. $7 - 22 = -15^{\circ}\text{C}$
- D. $22 \div 7 = 3.1^{\circ}\text{C}$

Correct answer: A. $22 - 7 = 15^{\circ}\text{C}$

Detailed answer:

Dew point depression is calculated by subtracting the dew point from the temperature:

$$22^{\circ}\text{C} - 7^{\circ}\text{C} = 15^{\circ}\text{C}.$$

This calculated value is then used on the chart's dew point depression scale.

17. Free State cloudy day

A Free State training flight has:

- Temperature: 17°C
- Dew point: 13°C

Which answer gives the correct dew point depression and best next action?

- A. 4°C; plot it with 17°C on the chart
- B. 13°C; ignore the chart
- C. 30°C; assume no icing
- D. 17°C; use only the right-hand humidity scale

Correct answer: A. 4°C; plot it with 17°C on the chart

Detailed answer:

$$17^{\circ}\text{C} - 13^{\circ}\text{C} = 4^{\circ}\text{C}.$$

The next correct step is to plot 17°C temperature and 4°C dew point depression and identify the shaded risk area at the intersection. The right-hand scale gives relative humidity reference, but the chart's main plotting process uses temperature and dew point depression.

18. Drakensberg foothills cool inland conditions

A pilot near the Drakensberg foothills records:

- Temperature: 10°C
- Dew point: 6°C

What is the dew point depression?

- A. 4°C
- B. 6°C
- C. 10°C
- D. 16°C

Correct answer: A. 4°C

Detailed answer:

$$10^{\circ}\text{C} - 6^{\circ}\text{C} = 4^{\circ}\text{C}.$$

The inland plateau and higher terrain can be cooler than coastal regions, and South Africa's interior is influenced strongly by altitude.

For carburettor icing assessment, the chart still requires the same method: temperature, dew point depression, intersection, shaded zone.

19. Pretoria warm inland afternoon

Pretoria conditions:

- Temperature: 29°C
- Dew point: 16°C

What is the dew point depression?

- A. 13°C
- B. 16°C
- C. 29°C
- D. 45°C

Correct answer: A. 13°C

Detailed answer:

$$29^{\circ}\text{C} - 16^{\circ}\text{C} = 13^{\circ}\text{C}.$$

This value should be plotted on the dew point depression axis against 29°C on the temperature axis.

The shaded area at the intersection gives the probability/severity category.

20. Inland comparison question

Which inland scenario has the highest moisture content relative to temperature, based on the smallest dew point depression?

- A. Johannesburg: 27°C / 12°C
- B. Kimberley: 34°C / 8°C
- C. Mbombela: 25°C / 20°C
- D. Upington: 36°C / 10°C

Correct answer: C. Mbombela: 25°C / 20°C

Detailed answer:

Calculate each:

- Johannesburg: $27 - 12 = 15^{\circ}\text{C}$
- Kimberley: $34 - 8 = 26^{\circ}\text{C}$
- Mbombela: $25 - 20 = 5^{\circ}\text{C}$
- Upington: $36 - 10 = 26^{\circ}\text{C}$

The smallest dew point depression is 5°C, so the Mbombela scenario has the moistest air among the choices. This is important because the chart uses dew point depression directly to determine the icing-probability zone.
